MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) REGULAR

EXAMINATION THIRD SEMESTER

MG3SDCREM202- AERODYNAMICS FOR WIND TURBINE GENERATORS

(2025 ADMISSION ONWARDS)

Duration: 1.5 hour **Maximum Marks:50**

Rememder(K) , ***Understand(U), Apply(A), Analyse(An),*** **Evaluate(E), Create(C), Skill(S),** ***Interest(I) and Appreciation(Ap)***

PART A

Short answer

Answer any 15 questions

Each question carries 1 marks

1. Define wind energy. [CO1 – U]
2. What is the kinetic energy of wind? [CO1 – U]
3. Define Tip Speed Ratio (TSR). [CO1 – U]
4. State the Betz limit. [CO1 – U]
5. What is meant by a Horizontal Axis Wind Turbine (HAWT)? [CO1 – U]
6. What is an airfoil? [CO2 – U]
7. Define lift force. [CO2 – U]
8. What is Reynolds number? [CO2 – U]
9. What is drag force?[CO3 – U]
10. Define wind turbulence. [CO3 –U]
11. What is wind shear?[CO3 – U]
12. Name any two instruments used for measuring wind speed and direction. [CO4 – U]
13. What is a wind rose? [CO4 – U]
14. Define Weibull distribution. [CO4 – U]
15. What is a power curve of a wind turbine? [CO4 – U]

16. Define capacity factor. [CO4 – U]

17. What is yaw control in a wind turbine? [CO4 – U]

PART B

Short Essay
Answer any 3 questions
Each question carries 5 marks

18. Explain the basic principle of wind energy conversion and derive the expression for the kinetic energy of wind. [CO1 – U]
19. Explain the different types of wind turbines with suitable examples. [CO1 – U]
20. Explain the aerodynamic forces acting on an airfoil. Differentiate between lift and drag. [CO2 – An]
21. Explain the Coriolis Effect, local effects, wind shear and turbulence in wind regimes. [CO3 – An]
22. Explain Weibull distribution and its importance in wind data analysis. [CO4 – An]

PART C

Essay
Answer any 2 questions
Each question carries 10 marks

23. a)Explain the working principle of a Horizontal Axis Wind Turbine with a neat schematic representation
    b) Explain Tip Speed Ratio and Betz limit and discuss their importance in wind turbine performance. [CO1 – U]

24. a) Explain the fundamentals of fluid mechanics applicable to wind turbines

    b) Explain airfoil geometry, lift and drag characteristics, Reynolds number effects and their importance in wind turbine blade design. [CO2 – A]

25. a) Explain the methods of wind measurement and classification of wind resources

    b) Discuss wind rose, wind speed distribution and statistical models used for wind data analysis. [CO3 – An]

26. a) Explain the power curve and performance characteristics of a wind turbine.

    b) Discuss pitch control, yaw control and stall-regulated versus pitch-regulated wind turbines. [CO4 – An]

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